Immunomodulatory Effects of Extracorporeal Photopheresis (ECP)

Extracorporeal photopheresis (ECP) exerts immunomodulatory effects through multiple pathways. Pro-inflammatory cytokines such as **tumour necrosis factor (TNF)-α** and **interleukin (IL)-6** induce the activation of **CD36-positive macrophages**. Notably, long-term immunological alterations can be induced by continuous ECP. In cutaneous T-cell lymphoma (CTCL), disease severity is associated with a Th1/Th2 immune imbalance—characterized by increased **IL-4** and **IL-5** production, reduced natural killer (NK) cell activity, and diminished cytotoxicity of **CD8-positive T cells**.

In a study of patients with early-stage CTCL (stage IB) undergoing ECP for 1 year, Di Renzo et al. observed not only an increase in **CD36-positive monocytes** in peripheral blood but also a shift in the cytokine profile of peripheral blood lymphocytes following phytohaemagglutinin stimulation. This suggests that ECP reverses the pathological Th2 skewing in CTCL and restores the **Th1/Th2 balance**. Additionally, ECP appears to promote the release of **anti-inflammatory cytokines** while reducing levels of **pro-inflammatory cytokines**.

ECP in GVHD, Transplant Rejection, and Autoimmune Diseases

Over time, ECP has demonstrated clinical benefit in patients with **graft-versus-host disease (GVHD)**, **transplant rejection**, and various **autoimmune diseases**. While the mechanisms observed in CTCL do not fully explain ECP's efficacy in these conditions, its therapeutic effects are thought to involve immune regulation rather than generalized immunosuppression—especially since ECP is not associated with increased risk of opportunistic infections.

In patients with chronic GVHD (cGVHD), ECP has been shown to induce **IL-10** production via modulation of arginine metabolism. The induction of

regulatory T (Treg) cells is a key proposed mechanism. Studies in murine models of contact hypersensitivity have demonstrated that an "ECP-like" procedure—using in vitro 8-methoxypsoralen (8-MOP) and UVA-exposed leukocytes followed by intravenous reinfusion—leads to the generation of Treg cells. These cells resemble UVB-induced Tregs and typically express **CD4**, **CD25**, **CTLA-4**, and the transcription factor **Foxp3**, and they suppress effector lymphocyte activity. **IL-10** release is also implicated in this regulatory process.

A recent clinical study involving 46 patients with cGVHD found that serum **B-cell activating factor (BAFF)** levels measured one month after ECP initiation could predict skin response at 3 and 6 months. Patients with BAFF levels < **4 ng/mL** showed significant improvement in skin symptoms.

Role of Treg Cells in ECP-Mediated Tolerance

Acute GVHD (aGVHD) in allogeneic transplant recipients is often linked to low numbers of Treg cells. ECP has been shown to induce T cells with regulatory properties in murine GVHD models. Multiple studies in both CTCL and GVHD patients report an **increase in Treg-cell numbers** and **enhanced suppressive function** following ECP treatment, providing a plausible explanation for its efficacy in immune-mediated conditions.

However, the role of Tregs in **Sézary syndrome (SS)**—a leukemic variant of CTCL—appears more complex. Patients with SS exhibit **reduced Treg-cell counts** and **impaired suppressive function**, raising questions about whether Tregs can effectively control malignant **CD4-positive T cells** in this context. This remains an area of ongoing investigation.

In lung transplant recipients, a recent study showed that ECP **slightly increased or stabilized** peripheral **$CD4^+CD25^+FoxP3^+$ Treg-cell counts**, particularly in those with stabilized graft function. Overall, the reinfusion of photochemically treated leukocytes mediates **antigen-specific suppression** of both humoral and cellular immune responses, promoting **allograft tolerance** and prolonging survival of transplanted organs.

The mechanism by which ECP counteracts cardiac rejection has been explored in murine models, further supporting its role in inducing immune tolerance through regulatory cell populations.